

Lab 3: Galaxy Image Classification and the Galaxy Merger Rate

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Abstract

In this lab, I developed a convolutional neural network (CNN) using PyTorch Lightning to morphologically classify images of galaxies from the Sloan Digital Sky Survey. The model was trained on crowd-sourced classifications from the Galaxy Zoo 2 (GZ2) project. It learned to predict features from 37 labels provided by Willet et al. 2013. The trained model was applied to an unclassified test set to estimate the fraction of galaxies currently undergoing major mergers. The inferred merger fraction was compared to theoretical predictions and observed fractions established by Lotz et al. 2011.

1 Introduction

For thousands of years astronomers relied on the eye to make observations. Growth in the field was stunted by the time it took for the Earth to revolve around the Sun. The sky was our limit for the human eye could not see past it. For these reasons, Hipparchus would have a heart attack if he knew the pitfalls modern astronomers face.

Four hundred years ago, as Hans Lipershay peered through the lens of the first telescope, the field of astronomy saw a paradigm shift. Progress was no longer halted by the tedium of tracking individual stars in the sky. Humans conquered the limitations imposed on our vision by Mother Nature, God or evolution, whichever you prefer. Over the last century, astronomers have been assaulted with data. We have observed with fervor, watching stars die the way one watches paint dry. Our insatiable appetite for understanding has encountered a new terminus; there are not enough humans to parse the data. Scientists have combated this by seeking assistance from the public. Willet et. al. 2013. recruited "Citizen Scientists" to morphologically classify galaxies. More sets of eyes allows for more classifications, however, it also results in more errors. Humans are susceptible to the pitfalls of bias, second guessing, and ignorance (myself included)!

Just as the telescope revolutionized archaic astronomy, CNN's are here to redefine contemporary astronomy. This new technology has the ability to automate the classification process with remarkable accuracy. In this lab, I will utilize the GZ2 dataset to train a deep learning model. The ultimate goal is to leverage this model to reliably estimate the local ($z \approx 0$) galaxy merger rate and compare it against standard cosmological models.

2 Methods

2.1 Data Processing and Label Analysis

The training dataset consisted of 61,578 color images of galaxies. The original images were downloaded at a resolution of $424 \times 424 \times 3$ pixels.

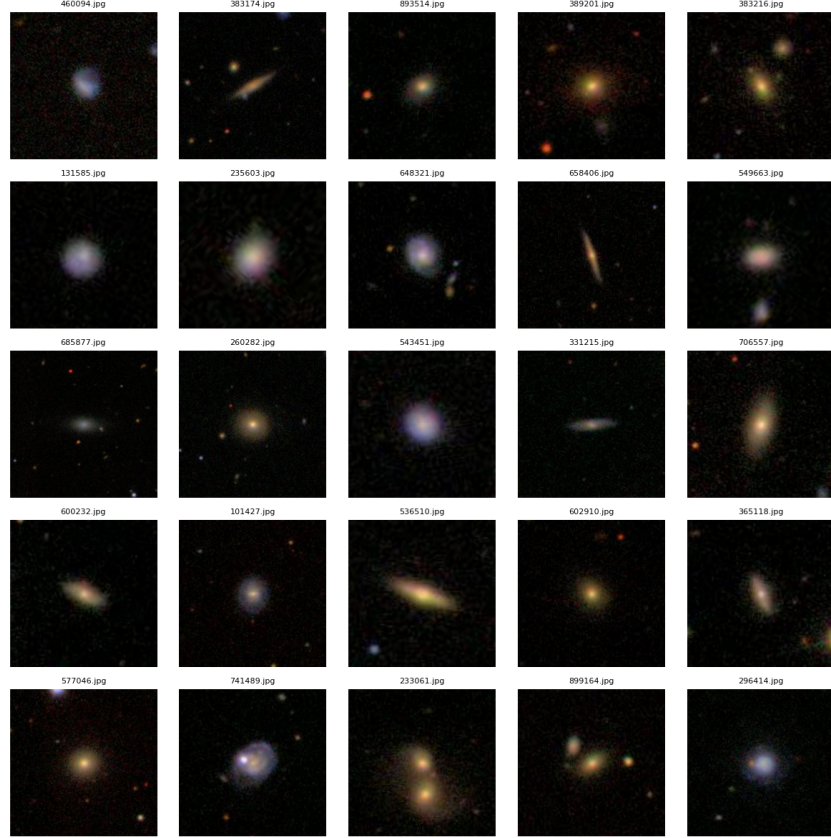


Figure 1: Twenty-five random images selected from the training data set (pre-downsizing).

The data set included a .csv file with the probabilistic label distributions. The histogram below plots the label distributions. The plot gives insight into which labels are easy to classify and which are ambiguous. Strong user agreement is indicated by high probability density around 0 and 1, while an even probability spread tells us that there was high disagreement for that label.

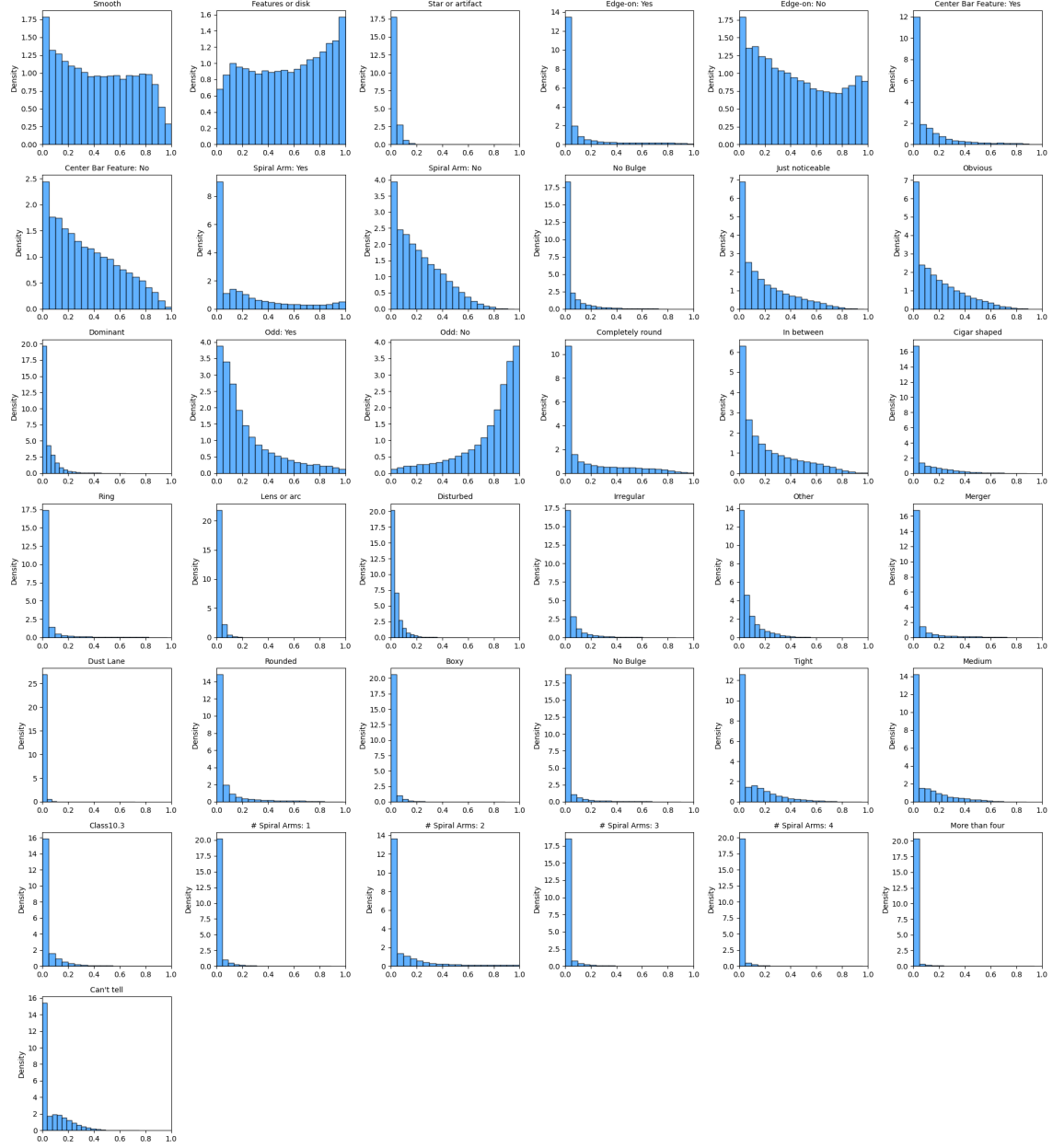


Figure 2: Histogram of the label distributions provided in training_classifications.csv

The labels with highest disagreement are: smooth vs. features or disk, edge on (yes/no), center bar feature, odd, and spiral arms (yes/no). For a detailed table of the classifications and the decision tree, refer to [A.1](#) in the Appendix.

A correlation matrix was constructed to assess the covariance between labels. One would expect

mutually exclusive labels to have anti-correlation, with a coefficient of -1. This is because they cannot be selected at the same time. However, there is only one label with a coefficient of -1. This is because of the user disagreement and ambiguity during the classification process.

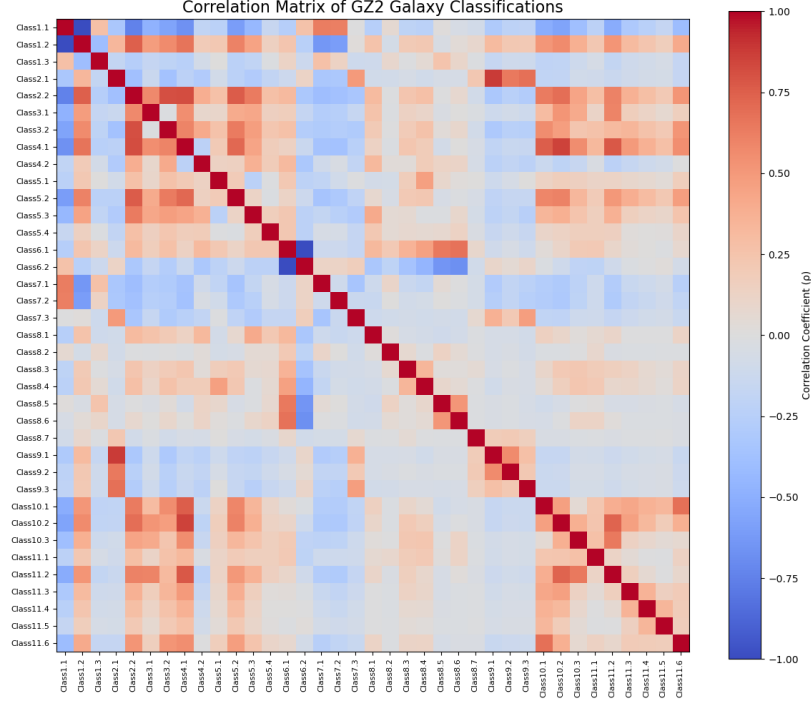


Figure 3: Correlation matrix heatmap that exemplifies relation between labels. A darker red indicates a strong positive correlation, while a dark blue indicates the opposite.

The heatmap communicates a lot of data, but the trade-off is that it is overwhelming. It was helpful to print the classes that had the highest magnitude coefficients.

Table 1: Top 15 Most Significant GZ2 Class Correlations

Label 1	Label 2	Correlation Coefficient
Class6.1	Class6.2	-1.0000
Class1.1	Class1.2	-0.9924
Class2.1	Class9.1	0.8785
Class4.1	Class10.2	0.8581
Class2.2	Class4.1	0.8139
Class2.2	Class3.2	0.8094
Class4.1	Class11.2	0.7811
Class2.2	Class5.2	0.7589
Class4.1	Class10.1	0.7545
Class1.2	Class2.2	0.7478
Class1.1	Class2.2	-0.7464
Class10.2	Class11.2	0.7309
Class4.1	Class5.2	0.7134
Class2.2	Class10.2	0.6883
Class2.1	Class9.3	0.6823

A.1 in the Appendix describes each classes value. The strong correlation between 2.1 & 9.1 is intriguing. This implies that edge on view is related to a center bulge. Such a result reveals a property of galaxy geometry! This morphological trait is likely due to the conservation of angular momentum forcing the rotating galaxy to spread out into a plane as it spins. An analogy would be pizza dough expanding radially outwards as it is tossed and spun.

4.1 & 10.2 was another fun one to note. Spiral arm patterns and medium tightness. This relationship reveals a weakness in the classifications. This is a subjective trait that is open to cognitive bias. Does the number of arms impact how tight one perceives the galaxy? Where is the boundary between tight, medium, and loose? This is not a standardized metric which leaves the data set open to influence from bias. It is unscientific and concerning because this bias will be reflected in the CNN trained on the data set.

A back of the napkin calculation to estimate memory allocation is as follows. The training images are 424×424 pixels. An RGB pixel takes 3 bytes of storage. If we use RAW images for the training, with a training set of 60000 images the minimum memory required is:

$$424^2 \times 3 \times 60000 = 3.24 \times 10^{10} \text{ bytes.}$$

If the images are stored as float 32, each pixel requires an additional 4 bytes of storage to accommodate decimal places values. The calculation becomes:

$$424^2 \times 3 \times 4 \times 60000 = 1.30 \times 10^{11} \text{ bytes.}$$

To accommodate memory constraints the images were center-cropped. This removed empty sky background. The images were then down-sampled to a resolution of 77×77 pixels. The data was split into an 80% training set and a 20% validation set.

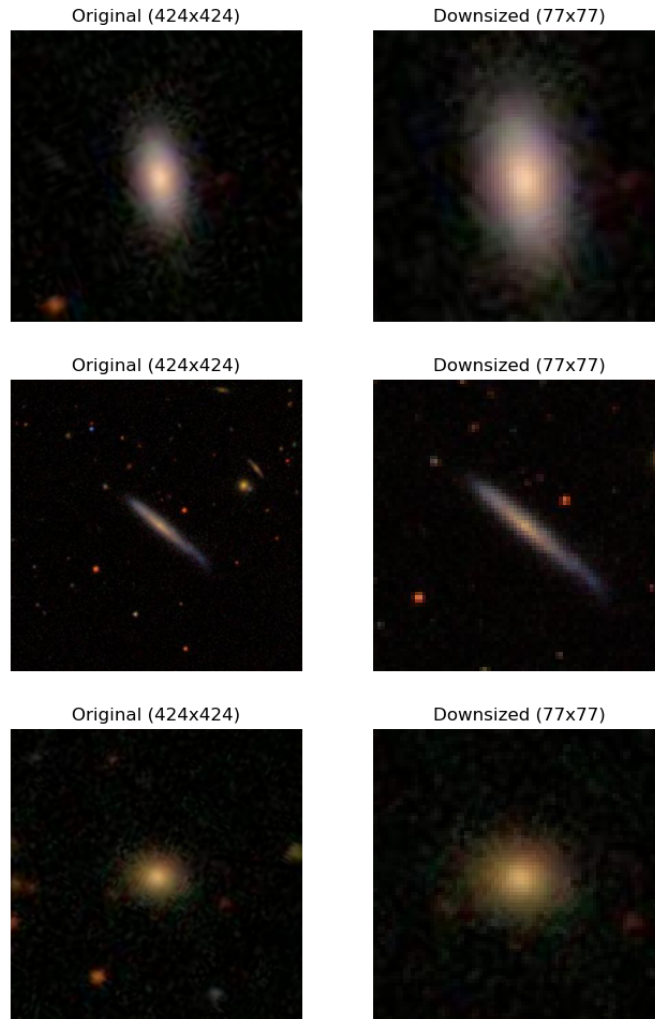


Figure 4: A comparison of test images before and after down-sizing

2.2 The Loss Function & ResNets

A loss function is necessary for building a neural network. It measures how well the model is predicting a desired output. The Root Mean Squared Error (RMSE) was the loss function used for the deep learning networks in this lab. A lower bound RMSE was calculated with a primitive model that assumes the labels for each image are the mean value of the corresponding label in the training set. The training set and validation set had RMSE's of 0.1638 and 0.1639, respectively.

Prior to building my own CNN, I utilized a ResNet architecture from PyTorch. A ResNet is a type of CNN that can accommodate complex and deeply connected networks without suffering from performance drops.

Classic CNN's suffer from performance drops when too many layers are added. This is because the mathematical model used to train the model becomes heavily altered and diluted if it has to interact with too many layers. The addition of too many connections hinders the accuracy!

ResNets sidestep this issue by skipping some of the connections. Sometimes it is best for a layer to leave an input unchanged, to avoid muddling the model too much. A CNN would do this by adjusting its weights to approximate the input as closely as possible. A ResNet just removes these layers. This is effective because it is easier for a network to push weights to zero than to learn a perfect identity transformation from scratch.

2.3 Designing a Custom Convolutional Neural Network

The custom CNN consists of four convolutional blocks followed by three fully connected layers. Each convolutional block applies a Conv2d layer (with padding=1 to preserve spatial size). This is followed by BatchNorm2d for training stability. Then a ReLU activation, and a MaxPool2d to halve the spatial dimensions. The channel depth doubles across blocks. The initial channel depth is 32 and it ends at 256. At the same time, the spatial size shrinks, descending as follows: 77, 38, 19, 9, 4. After the final convolutional block, a global average pool collapses each feature map to a single value. This produces a 256-element vector. The vector is fed into three fully connected layers with ReLU activations and a Dropout of 0.5 between them. The Dropout and ReLU activations are intended to combat overfitting. The final layer uses a Sigmoid activation to constrain all 37 outputs to $[0, 1]$. This matches the probability-like nature of the GZ2 labels. The optimizer is Adam with an initial learning rate of 1×10^{-3} .

The custom CNN contains 661,797 free parameters. The training set has a pixel count of 876,223,194. This makes me somewhat nervous. A model with more free parameters than training pixels can in principle memorize the training set rather than learning generalizable features. However, two factors mitigate this. First, each pixel is just one number, while each parameter is reused across every image in a convolutional layer. As a result, the effective ratio is more favorable than it looks. Second, BatchNorm and Dropout actively constrain what the model can memorize.

The ResNet contains 11,187,813 free parameters. Despite having far fewer parameters, my custom CNN can still perform competitively because global average pooling keeps the fully connected section small. The ResNet compensates for its larger size with residual connections that make it easier to train deep networks without overfitting.

2.4 Scheduler Implementation

After comparing the custom CNN to the ResNet, it was determined that the ResNet performed better. A learning rate scheduler was introduced to adjust the step size of the optimizer during training. This is useful for fine tuning the accuracy of the model. As the model converges toward the global minimum, large steps become a detriment. They cause the optimizer to overshoot the absolute minimum. A scheduler progressively lowers the learning rate. This causes the optimizer to take more precise steps. This allows the model to explore the steepest regions of the minimum without bouncing out. Doing this rewards the model with a lower final loss and higher classification accuracy.

3 Results

3.1 The PyTorch ResNet

The first ResNet architecture I implemented was the most basic. The network was trained from scratch with randomly initialized weights; whereas the later models utilized transfer learning by loading pre-trained weights. I ran the model for 15 epochs. The best validation RMSE was 0.0950. It occurred on the final epoch.

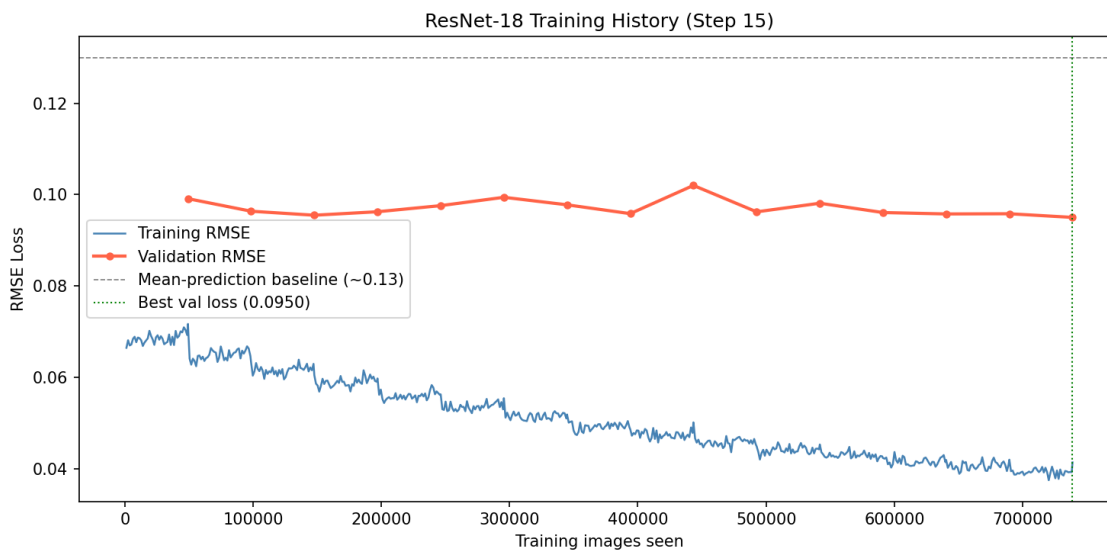


Figure 5: Training and validation loss of the most basic ResNet architecture implemented. After 15 epochs the best validation loss was 0.0950.

3.2 Custom CNN



Figure 6: Custom CNN loss plot. The best validation RMSE was 0.0978. It underperformed when compared to the ResNet.

3.3 ResNet with Scheduler

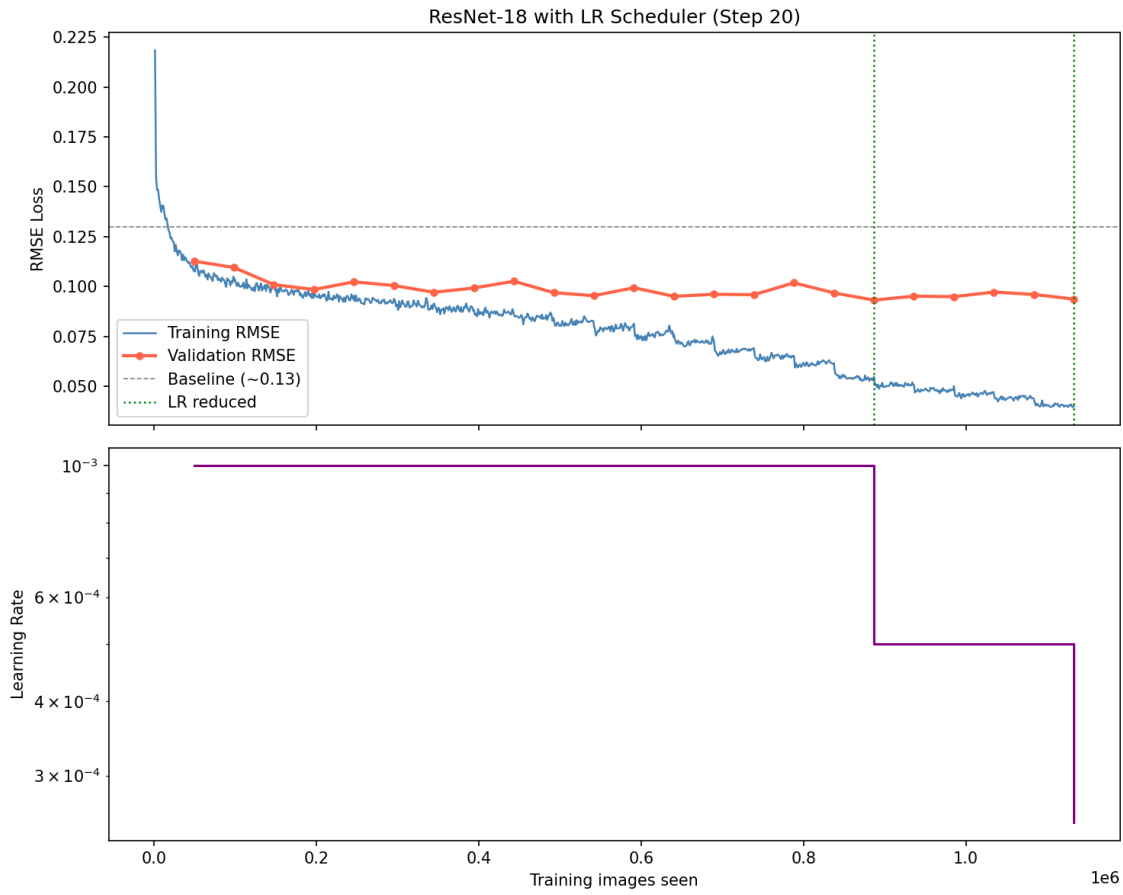


Figure 7: RMSE loss with the scheduled ResNet. The models loss improved. The lowest RMSE occurred on epoch 18 with a value of 0.0932

3.4 ResNet with Scheduler & Data Augmentation

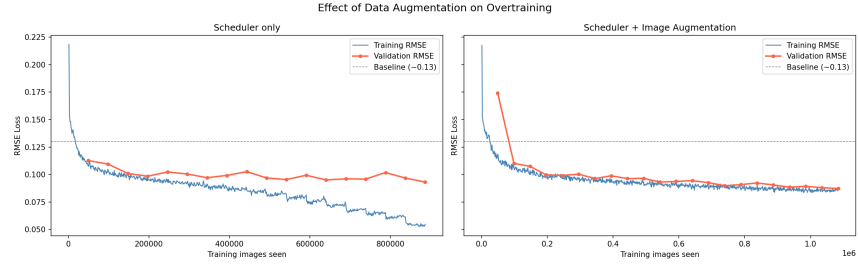


Figure 8: Comparison of ResNet before and after data augmentation

3.5 Loss Comparison Across All Models

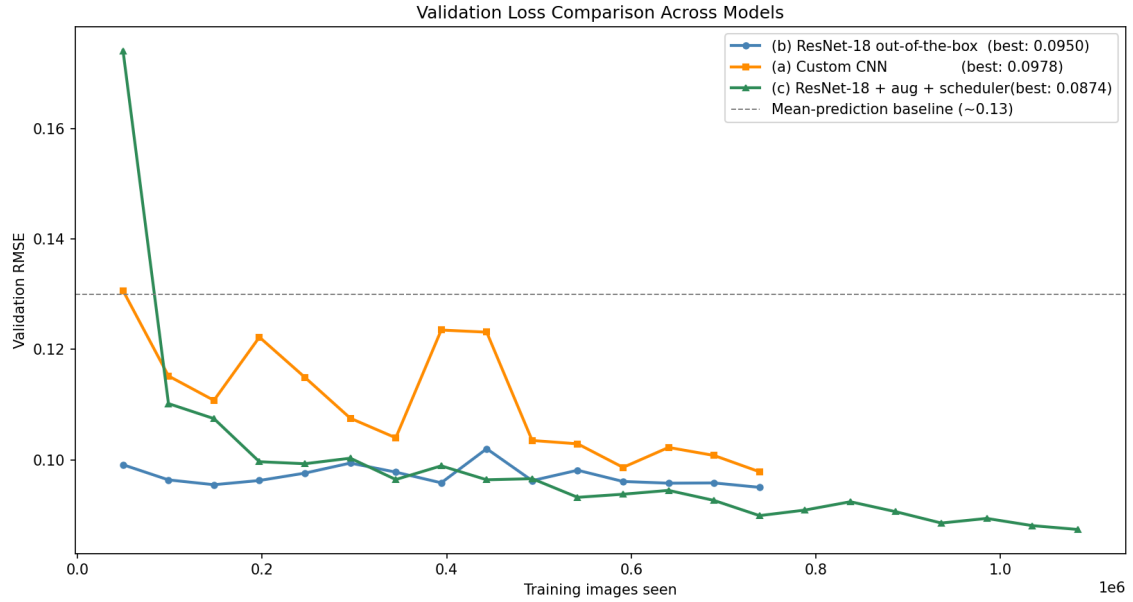


Figure 9: A comparison of RMSE loss across all of the models. The ResNet that utilized the scheduler and data augmentation performed the best with an RMSE of 0.0874

3.6 Model Performance Across Labels

To evaluate the performance of the convolutional neural network, I plotted the RMSE for all 37 classes. Overall, the model performs well. A majority of the classification labels had an RMSE below 0.13. This indicates that for most morphological features, the model's predicted probability is within 15% of the human consensus vote fraction. It also means that it performed better than

the baseline model from the start of the lab.

The model exhibited the highest error rates for the first branching question in the GZ2 decision tree. This was the choice between "Smooth" and "Features or Disk". This presents as an issue because this decision directly affects every subsequent label that the model evaluates for that galaxy. A wrong choice in this label immediately skews the validity of the rest of the labels. The reason this label has such a high RMSE can be chalked up to the training set. The citizen scientists who evaluated the galaxies suffered from high levels of disagreement when evaluating these categories. It is likely that this error seeped into the model.

Despite this, the model yielded very low errors for quite a few of the labels. Variants within class 8 and 11 have the lowest RMSE. The values are clustered tightly near 0.01 to 0.05. The labels in these classes include: odd features such as rings, lenses, disturbed, or mergers, and the specific number of spiral arms.

3.7 Manually Assessing Validation Accuracy

In order to manually check how well the model was performing, I plotted the images of the 5 objects in the validation set that are the most extreme example of a given label. I did this for 7 of the labels: (1) smooth, (2) star/artifact, (3) edge-on disk, (4) odd: ring, (5) odd: lens/arc, (6) spiral: 2 arms, (7) odd: merger.

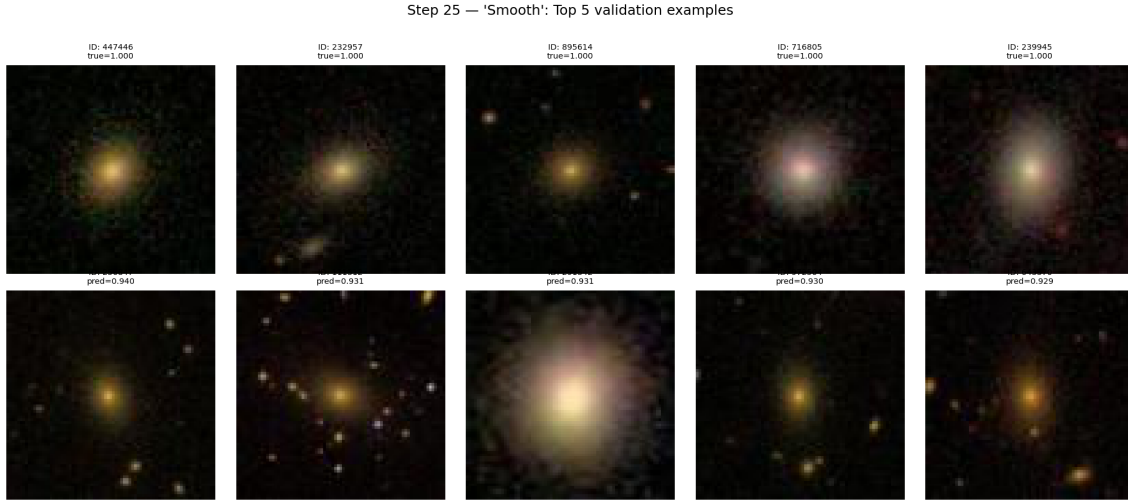


Figure 11: Model Validation Assessment: Smooth.

The model was reliable for this label. The images all follow the same form. Despite "smooth" being one of the most difficult labels for humans to classify, the model seems to have performed well.

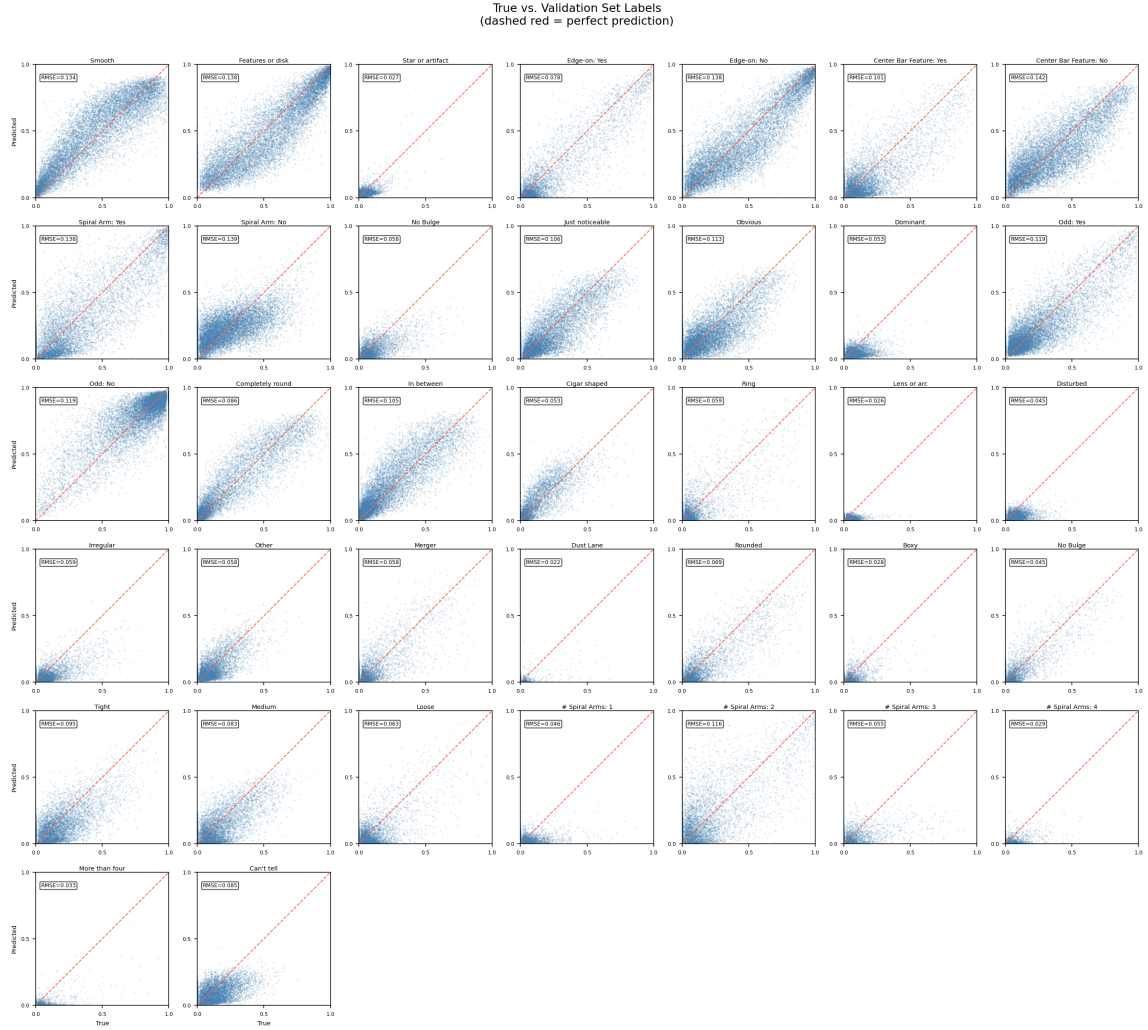


Figure 10: Scatter plot that indicates the models prediction performance for each label. A plot perfectly centered on the regression line would indicate perfect performance.

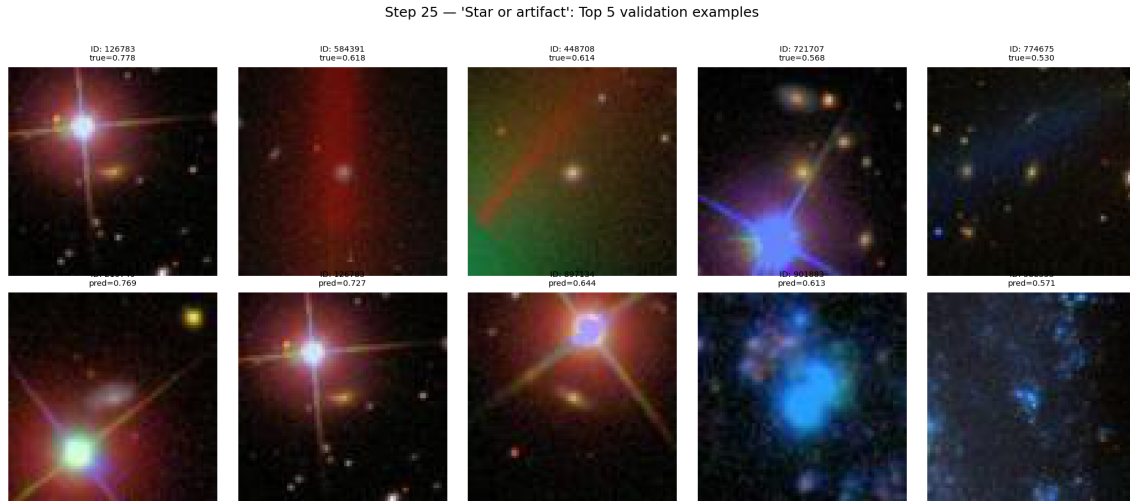


Figure 12: Model Validation Assessment: Star/Artifact

The model was able to accurately identify stars/artifacts. This label is easier to distinguish than others, so I am not shocked with the results.

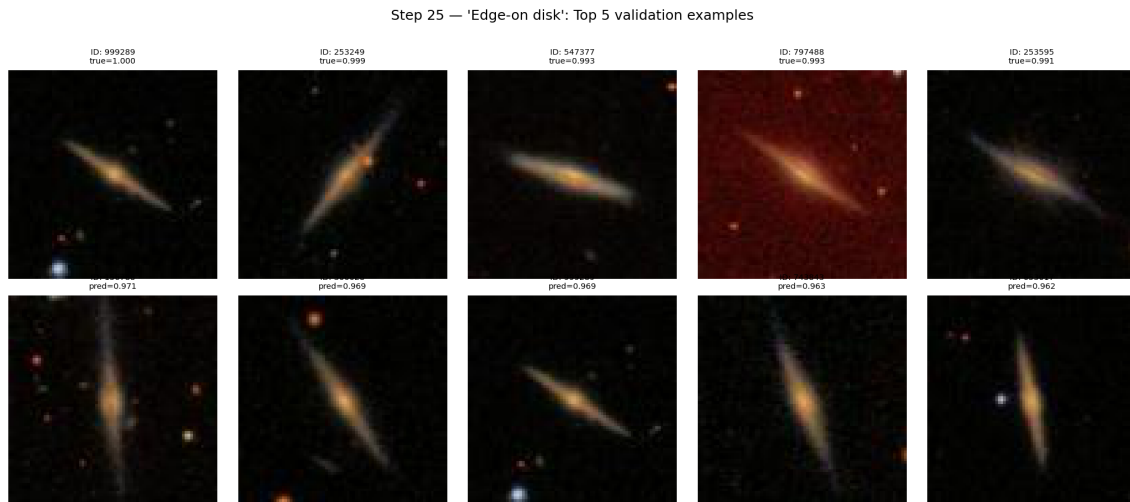


Figure 13: Model Validation Assessment: Edge-on disk.

The model's predictions match the validation set.

The model's predictions seem reasonable. As a human, I got this label confused as a disk artifact. However, the model is confident in these selections, and the RMSE of this particular label is low, 0.059.

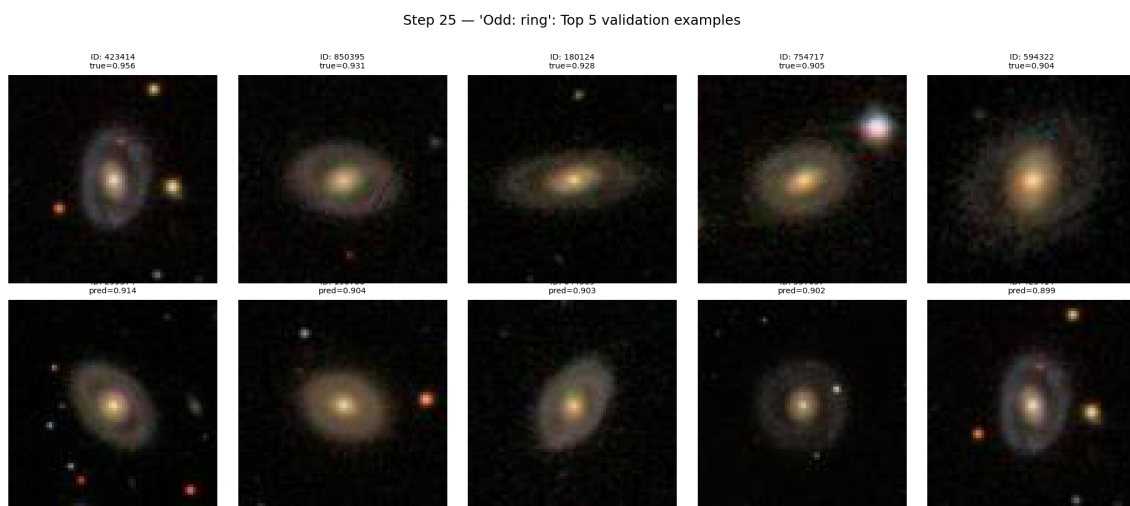


Figure 14: Model Validation Assessment: Odd: Ring.

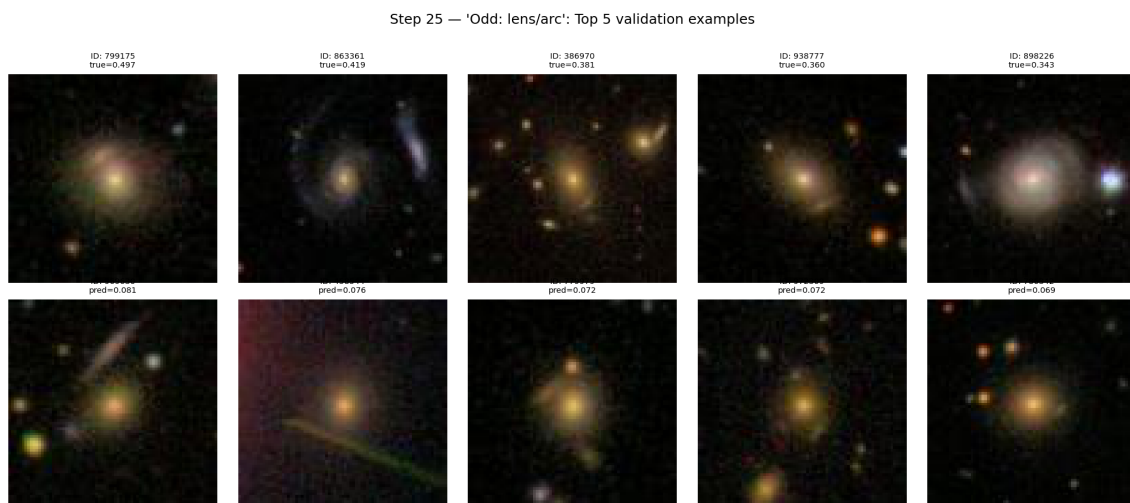


Figure 15: Model Validation Assessment: Odd: Lens/Arc.

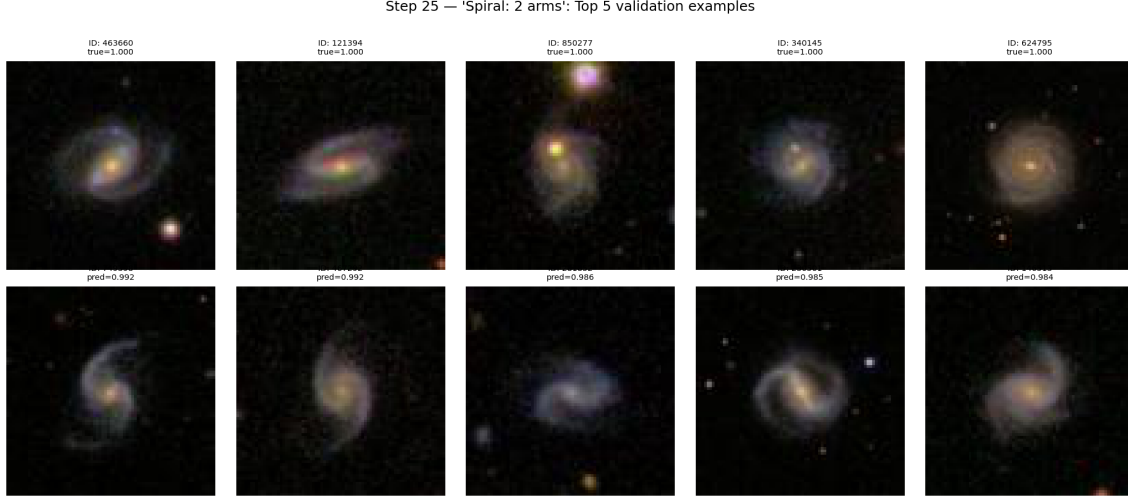


Figure 16: Model Validation Assessment: Spiral Arms: 2

The model underperformed on this label. Not all of the sample images have glaringly obvious lenses or arcs. While the model made good decisions on some of the images, but a couple lack the presence of a lens or arc entirely. I was shocked to see that this label had one of the lowest RMSE's of the entire set of labels. This could be due to the fact that users poorly classified this label.

For this label, the model seems to have outperformed the humans! Two of the example images with the highest true label value do not seem to possess two spiral arms at all. However, all of the labels that the model selected are clear cut examples of galaxies with two spiral arms. This result is amazing! How is it possible that the model can identify a label well, when the label was poorly classified by the users whos decisions ultimately served as the training data for the model?

This label threw me for a loop. Initially I thought I pulled the edge on: yes label twice. None of the images in the validation set seem to show mergers. Naturally the model is then trained to misunderstand what a merger actually looks like. It is worth noting that the odd dataset is small, which means that it is unreliable. The probability that the true classifications are mergers are low as well. A majority of the humans disagreed on the classifications. The model seems to doubt this too, it has very low confidence scores for each image selected.

3.8 Merger Fraction Estimation

The trained model was applied to the full test image set to estimate the galaxy merger fraction at low redshift. For each galaxy the model predicted a merger score. Galaxies with a predicted score above 0.5 were classified as mergers. This yielded a raw merger fraction of 1.3%. Visual inspection of the top 25 highest scoring galaxies confirms that the model is identifying genuine mergers. All 25 show classic merger signatures such as tidal tails, double nuclei, and interacting galaxy pairs. Despite the noisy GZ2 merger labels the model has learned meaningful morphological features associated with merging systems. The model seems to have outperformed the citizen scientists!

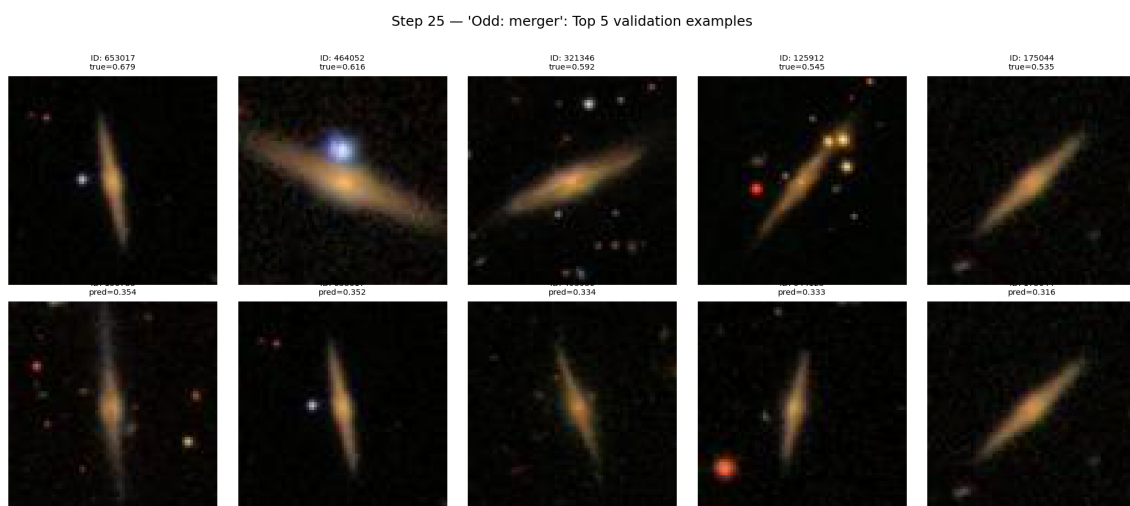


Figure 17: Model Validation Assessment: Odd: Merger

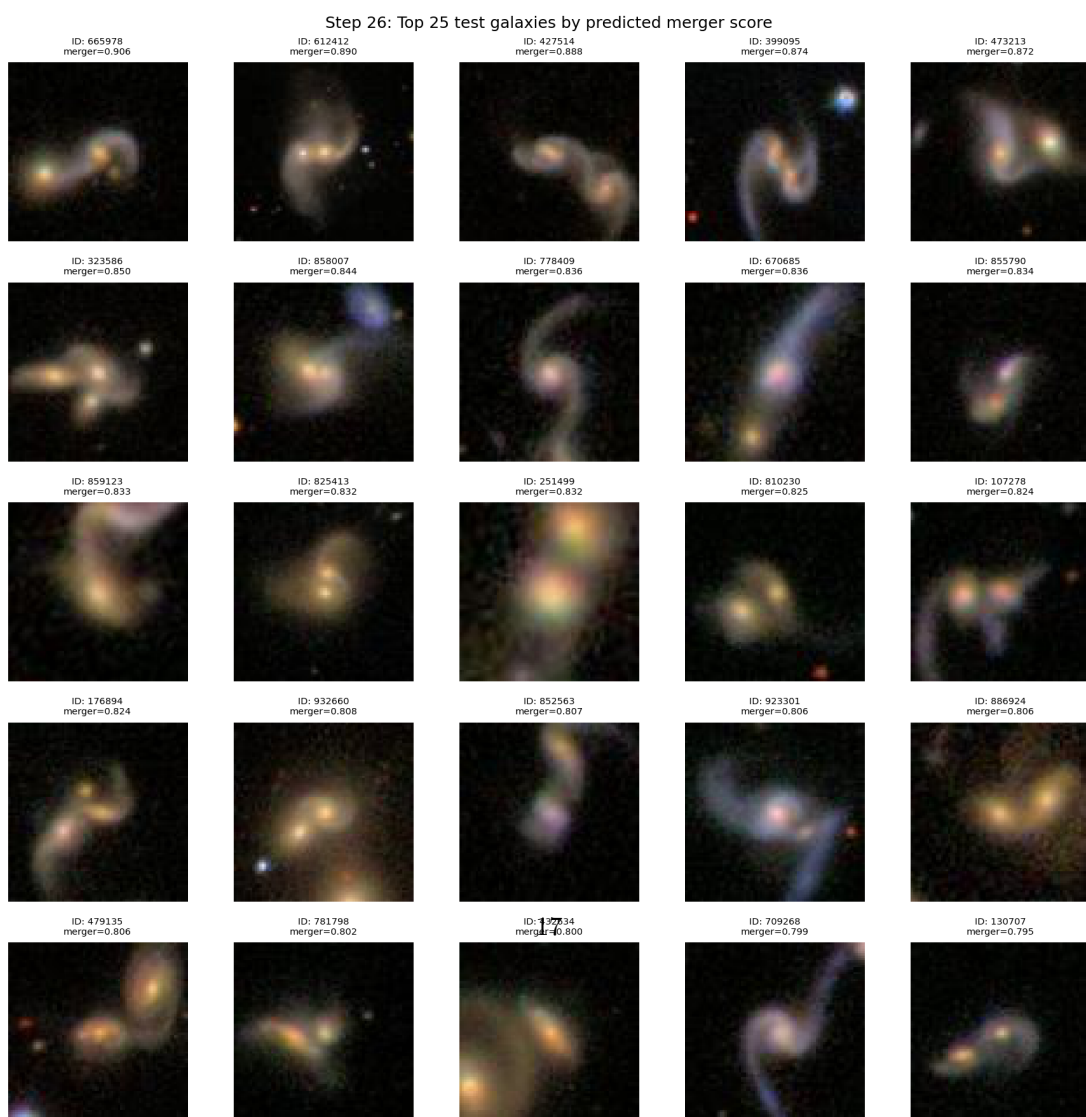


Figure 18: The top 25 Mergers predicted by the model. All of the images are clear examples of mergers, which indicates that the model performed very well!

The distribution of predicted merger scores is strongly peaked near zero. There is a long tail extending to high values. Few galaxies exceed the 0.5 threshold. This is a realistic result. Mergers are rare events at low redshift. The model seems to have learned to be conservative. This is probably because merger scores are almost always low in the training set. Predicting near zero would be the RMSE minimizing strategy.

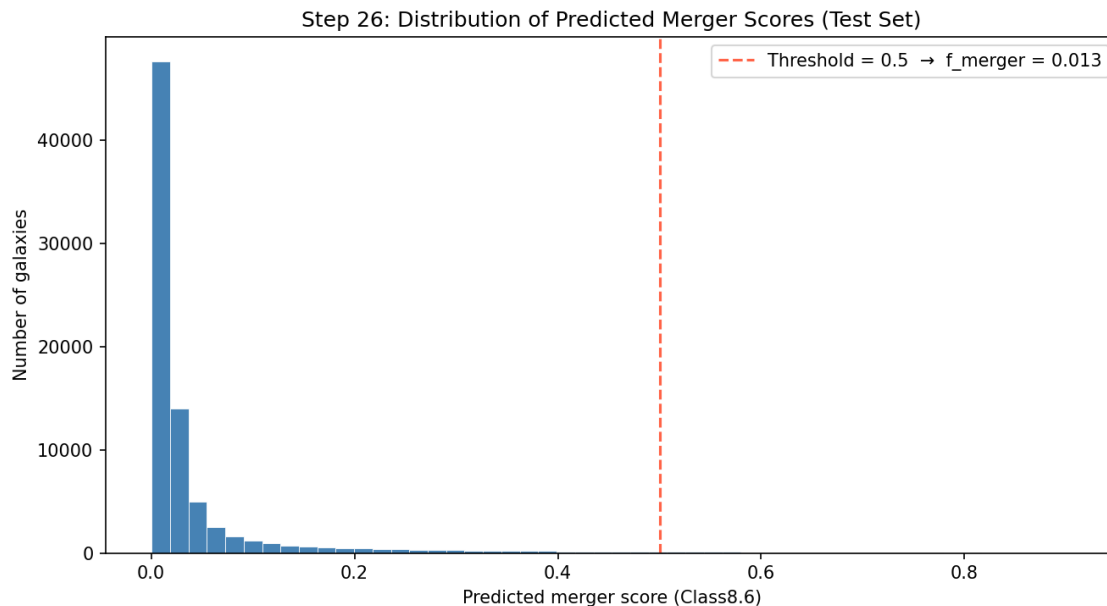


Figure 19: Model predicted galaxy merger scores.

4 Discussion

4.1 Galaxy Zoo Classifications

The training data for this lab comes from citizen scientist classifications. The first step was to invoke empathy for the slaves kept in Campbell Hall's basement, hunched over keyboards, classifying galaxy types. We did this by classifying some Galaxy Zoo JWST observations. This step did not instill confidence in me. Many of the images were grainy and difficult to see. The classifications were vague, and I often found myself questioning which label to select. A classification that bothered me read:

"How tightly wound do the spiral arms appear?"

1. tight
2. medium
3. loose

What does that mean?!?! This label felt unscientific and largely susceptible to user disagreement and error. What one person considers medium tightness could be wildly different from the next. The whole process of classifying these galaxies felt largely ambiguous, and it concerned me that this could be reflected in the data set that I used to train my model.



Figure 20: A 'smooth' galaxy from the Galaxy Zoo classification website

Figures 2 and 20 depict examples of the images from the Galaxy Zoo website. Note that these images were selected due to their above average resolution on the website.

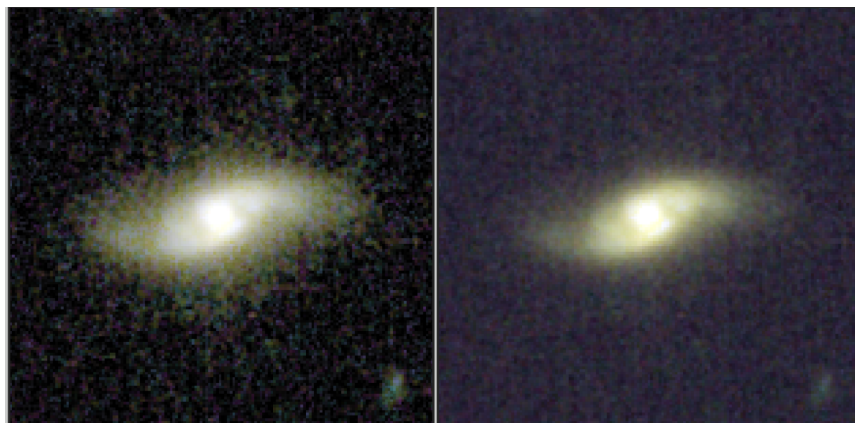


Figure 21: A 'spiral' galaxy from the Galaxy Zoo classification website. This galaxy has two spiral arms. Don't ask me how tightly wound they are.

4.2 Willet et al. 2013. Q & A

1. **What cuts were applied to the SDSS DR7 imaging survey to construct the Galaxy Zoo 2 (GZ2) sample? How many galaxies are in GZ2?**

Willet et al. aimed "to include only the nearest, brightest, and largest systems for which fine morphological features can be resolved and classified". They achieved this by only including galaxies with a Petrosian half-light magnitude brighter than 17.0 in the r -band (after Galactic extinction correction) and a size limit of $petroR90_r > 3$ arcseconds. Galaxies with a spectroscopic redshift outside the range of $0.0005 < z < 0.25$ were removed, as well as objects flagged by the SDSS pipeline as saturated, bright, or blended without an accompanying nodeblend flag. In total, the GZ2 project contains 304,122 galaxies.

2. **Describe the galaxy images in GZ2. What are the angular dimensions of the images? Are they the same for all galaxies? Roughly what physical scale in kpc does this correspond to?**

The GZ2 images are gri color composites. The images do not have the same angular size for all galaxies. Each image is 424×424 pixels, but is scaled dynamically to $(0.02 \times petroR90_r)$ arcseconds per pixel. The scaling relies on the Petrosian radius containing 90% of the r -band flux. As a result the physical scale of the image is adjusted to encompass the entire galaxy. The rescaling adjusts the physical scale to tens of kiloparsecs depending on the individual galaxy's size and redshift.

3. **Describe the classification scheme in GZ2. How many questions are asked for each galaxy?**

The GZ2 classification scheme uses a decision tree consisting of 11 classes with a total of 37 possible responses. The number of questions asked varies with the responses you choose.

4. **How did the classification scheme change from GZ1 to GZ2?**

GZ1 only asked for basic morphological distinctions (i.e. early-type ellipticals, late-type spirals, and mergers). GZ2 introduced a more complex decision tree to measure finer morpholog-

ical features. These new features include, but are not limited to, the presence of bars, bulges, the shapes of edge-on disks, arm curvature, and quantifying the relative strengths of galactic bulges and spiral arms.

5. How does GZ2 decide which galaxies to show a user and in what order?

At first, images were randomly selected from the database to be shown to classifiers. Toward the end of the project, images with a low number of classifications were shown at a higher rate to ensure every galaxy had enough responses to accurately characterize its classification likelihood.

6. How many unique classifications did a typical galaxy get (at the time this paper was written)?

The main sample galaxies had a median of 44 classifications per galaxy.

7. What steps were taken to deal with unreliable classifications?

Willet et al. employed an iterative weighting scheme to reduce the influence of unreliable classifiers. Unreliable classifiers were identified by comparing the choice to random selections. A user’s consistency was calculated by comparing their individual votes to the consensus vote fraction for each task. A weighting function was then applied to down-weight the classifiers who fell in the tail of low consistency.

8. What is the primary reason for apparent redshift evolution within the GZ2 sample?

The primary reason for apparent redshift evolution is classification bias. Because more distant galaxies appear smaller and dimmer in the cutout images, their finer morphological features become much more difficult for human classifiers to identify. Traits such as spiral arms and bars are especially susceptible to misclassifications.

9. Briefly describe (in words) how GZ2 attempts to correct for classification bias. What is meant by classification bias?

Classification bias is the phenomenon where the limited resolution and surface brightness of higher-redshift objects causes a user to select incorrect classifications. To correct this, GZ2 assumes that galaxies of a given absolute physical brightness and physical size should statistically share the same average mix of morphologies regardless of redshift. They compute the average morphology ratios for a local baseline at low redshift, and apply correction factors to the vote fractions of higher-redshift galaxies to match the unbiased baseline.

10. Why are higher-redshift galaxies in GZ2 more likely to be classified as mergers?

Due to angular separation bias, galaxies with a close apparent companion are often perceived differently by participants. The mean “merger” vote fraction increases strongly as angular separation decreases. The projected angular separation between non-interacting pairs of galaxies decreases at higher red shifts. This leads to an increased rate of projected pairs being classified as mergers.

11. GZ2 classifies each galaxy using boolean yes/no questions for a finite set of possible classifications. Yet, the reported classifications are all floats (typically ranging from 0 to 1). Why is this?

The reported classifications are reported per galaxy. The individual boolean responses from

all users are combined and averaged into a vote fraction for each class. This gives the probabilistic of that morphological feature. Therefore user disagreement results in a floating decimal between 0 and 1. If 7 out of 10 users agreed that a particular galaxy is viewed edge on, the classification for that galaxy would have a 0.70 probability of being "yes".

12. Summarize the most significant differences between GZ2 classifications and similar classifications from other morphological catalogs.

GZ2 includes significantly more systems than other catalogs. Additionally, each galaxy receives a large number of independent inspections, rather than just one expert's opinion. GZ2 provides continuous estimates of classification likelihoods and feature strengths, allowing for better statistical studies of rare objects and small-scale features.

4.3 The Importance of Morphological Classification

This lab is difficult and time consuming... What are the fruits of my labor? Casting aside innate curiosity, why do we care about galactic properties? Astronomers are interested in galaxy morphology because these features generate clues about the physics governing the systems! Exactly a century ago, Edwin Hubble proposed a scheme for classifying galactic evolution by morphology. This implies that we can infer a lot about a galaxies properties from its morphology.

Disturbed morphologies such as tidal tails, shells, asymmetric structures, etc. signify a galactic merger event. Classifying these irregular galaxies and measuring their frequency across different redshifts lets astronomers trace the cosmic merger rate. This information is helpful for testing the Λ CDM models of hierarchical structure formation.

The star formation rate and stellar population of a galaxy is strongly correlated to its morphology. Disk-dominated spiral and irregular galaxies possess cold gas reservoirs and exhibit active star formation. This tells us to look for these galaxies in the UV range! Bulge-dominated elliptical galaxies consist of older stellar populations. As a result elliptical galaxies are brightest in the IR spectrum. The morphological transition from disk to elliptical tells us about a galaxy's history.

Finer structural features like central bars, spiral arms, and rings provide evidence of slow internal evolution. Galactic bars funnel cold gas from the outer disk into the central regions. This process can fuel Active Galactic Nuclei. Knowing this allows us to study the dynamics of such systems. We can also learn how these phenomena alter the galaxy's long-term evolutionary trajectory.

4.4 Identifying Merging Galaxies

Merging galaxies represent a fundamental mechanism for mass assembly and structural transformation in the universe. The Λ CDM cosmological model offers a well accepted explanation for galaxy formation; small structures continuously merge to form progressively larger ones over long timescales. Therefore, the merger rate is essential for testing cosmological models.

Furthermore, major mergers drastically alter a galaxy's evolutionary trajectory. The immense gravitational torques produced during a merger can drive massive inflows of cold gas into the galactic center. This process can trigger intense bursts of star formation and funnel material into the central supermassive black hole.

Lotz et al. (2011) and Rodriguez-Gomez et al. (2015) describe the most popular methods for identifying mergers: morphological disturbances and close kinematic pairs.

Morphological features like tidal tails, stellar streams, shells, bridges, and double nuclei indicate a merger event. This makes visual classification an effective method of identifying mergers. To make this process scalable, astronomers developed non-parametric morphological indices. The CAS system (Concentration, Asymmetry, and Clumpiness) and the Gini- M_{20} parameters are the most popular. Galaxies undergoing active major mergers typically exhibit highly asymmetric light distributions and distinct Gini- M_{20} coefficients.

Unfortunately, morphological disturbances are only visible for a limited timescale. Additionally, high redshifts can dampen the tell tale features. As a result, astronomers also identify mergers by searching for close pairs. This technique involves identifying pairs of galaxies that share a very small projected physical separation on the sky (e.g., $\Delta R < 20 - 30$ kpc) and a small relative line-of-sight velocity difference (e.g., $\Delta v < 500$ km/s). These parameters indicate that the galaxies are gravitationally bound and destined to merge.

4.5 Comparing Results with Lotz et al.

Several factors complicate a direct comparison to Lotz et al. 2011. Their paper quote merger rates in units of Gyr^{-1} rather than dimensionless fractions. I divided my merger fraction by a typical disturbance timescale of 0.5 Gyr. This gave me an implied merger rate of $0.026 Gyr^{-1}$. This value falls within the observed range of $0.02-0.08 Gyr^{-1}$ shown in Lotz et al.’s Figure 13. Another issue is that Lotz et al. specifically study major mergers with mass ratios greater than 1:4. The GZ2 merger label makes no distinction between major and minor mergers. Therefore, my fraction likely includes some minor merger events. This may inflate the estimate relative to theirs.

As discussed in Willett et al. 2013, higher-redshift galaxies in GZ2 are more likely to receive merger votes. This is because poor angular resolution makes ordinary galaxies appear more morphologically disturbed. If the test set spans a range of redshifts this bias will inflate the merger fraction relative to the true $z = 0$ value.

Finally, the GZ2 merger label is weighted by the fraction of classifiers who first voted "odd". This means Class 8.6 represents a smaller value than the raw fraction of all classifiers who identified a merger. As a result, my estimate gets pushed downward. This helps compensate for the aforementioned biases, but it is still a systematic error that is worth noting.

Altogether, my estimated merger fraction of 1.3% is in reasonable agreement Lotz et al. 2014. The theoretically predicted and observed major merger fraction that they quote is 1-4% at $z = 0$.

5 Conclusion

This lab demonstrated the usefulness of convolutional neural networks to morphologically classify galaxies. In a matter of hours I was able to reproduce what would take human classifiers years to accomplish. I utilized Galaxy Zoo 2 labels as training data and built and compared three models. A custom CNN, an out-of-the-box ResNet-18, and a ResNet-18 with data augmentation and a

learning rate scheduler. Every model outperformed the mean prediction baseline of RMSE 0.13. The ResNet-18 trained with random rotation augmentation and a scheduler performed best. It achieved a validation RMSE below 0.09.

Model performance varied considerably across the 37 labels. Visually obvious features such as smooth morphology, edge-on disk orientation, and star/artifact classification were predicted reliably. Labels deeper in the GZ2 decision tree, particularly labels in class 8, were harder to classify. This is because they are rare in the training set and because these labels are difficult to identify.

The best model estimated a raw merger fraction of 1.3%. This result is consistent with the $z = 0$ major merger rates of 1–4% reported in Lotz et al. 2011. Visual inspection of the highest scoring merger candidates confirms that the model is identifying genuine mergers. It is worth noting that there are several systematic effects that complicate the comparison to theoretical predictions. These factors include the consolidation of major and minor mergers in the GZ2 label, redshift dependent classification bias, and the conservative predictions that result from training with an RMSE loss on a heavily imbalanced label.

Viewed from a broader lens, I think this lab highlights the power and limitations of deep learning models. From the get-go it became clear that model can only be as good as its training data. The innate noise within crowd sourced classifications places a ceiling on the models performance. No matter how much I tweak or train the model, there will always be a limit to what it can achieve. With access to cleaner labels, and more computing power/time, I could create a better performing model. However, with the restraints imposed, I am VERY pleased with my models performance. The results confirm that deep learning is a powerful tool for galaxy morphology classification. Buckle up astronomers!

A Appendix

A.1 Classification Tree from Willet et al. 2013

Task	Question	Responses	Next Task
01	Is the galaxy simply smooth and rounded, with no sign of a disk?	Smooth	07
		Features or disk	02
		Star or artifact	None
02	Could this be a disk viewed edge-on?	Yes	09
		No	03
03	Is there a sign of a bar feature through the centre of the galaxy?	Yes	04
		No	04
04	Is there any sign of a spiral arm pattern?	Yes	10
		No	05
05	How prominent is the central bulge, compared with the rest of the galaxy?	No bulge	06
		Just noticeable	06
		Obvious	06
		Dominant	06
06	Is there anything odd?	Yes	08
		No	None
07	How rounded is it?	Completely round	06
		In between	06
		Cigar-shaped	06
08	Is the odd feature a ring, or is the galaxy disturbed or irregular?	Ring	None
		Lens or arc	None
		Disturbed	None
		Irregular	None
		Other	None
		Merger	None
		Dust lane	None
09	Does the galaxy have a bulge at its centre? If so, what shape?	Rounded	06
		Boxy	06
		No bulge	06
10	How tightly wound do the spiral arms appear?	Tight	11
		Medium	11
		Loose	11
11	How many spiral arms are there?	1	05
		2	05
		3	05
		4	05
		More than 4	05
		Can't tell	05